

FORECASTING THE FUTURE OF MICROSOFT STOCK

D214 Capstone Presentation
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Introduction & Background

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Senior Database
Administrator

13 years of SQL

MSDA Student



PROBLEM STATEMENT

Stock prices fluctuate from day today making stock price prediction important to a company shareholders as nobody invests to lose money.

The objective is to analyze the historical stock price and identify any patterns or trends to develop a model that can accurately predict the future stock price.

Research Question:

Can we forecast the future stock price using time series analysis based on historical stock data?

HYPOTHESIS

Null Hypothesis

- The future stock price cannot be predicted with 80% accuracy

Alternate Hypothesis

- The future stock price can be predicted with 80% accuracy

DATA ANALYSIS PROCESS



DATA ANALYSIS PROCESS

Data Cleaning

Data Load

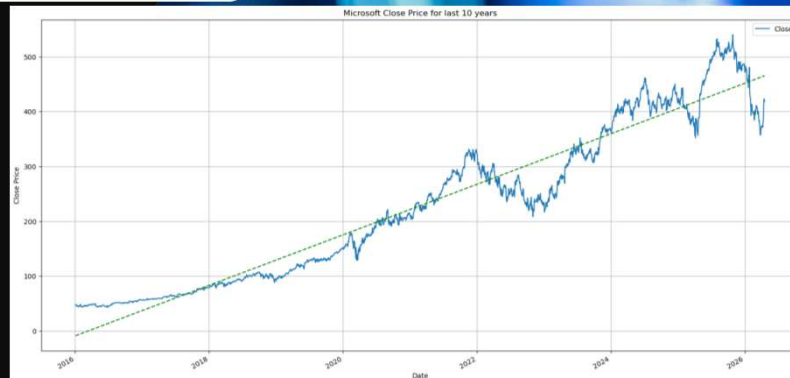
- Download Data from Kaggle
- Load Data into Pandas data frame

Explore Data

- Columns include Date, Close, High, Low, Open & Volume
- March 26th 1986 to April 20th 2026
- 10,094 rows

Filter Data for Model

- 10 years of data(January 1st 2016 to April 20th 2026)
- Date and Close columns
- Export new data frame



DATA ANALYSIS PROCESS

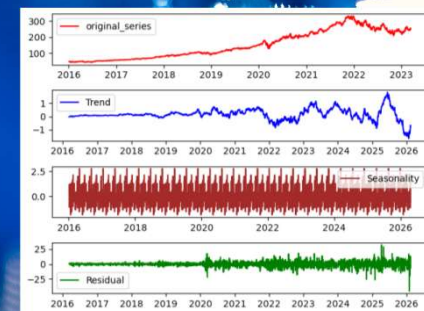
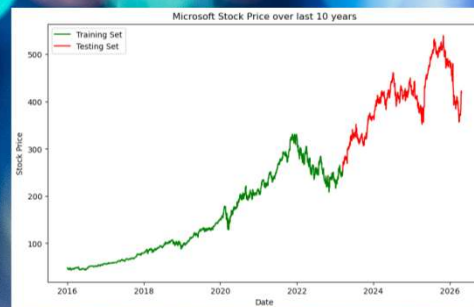
Data Preparation & Stationarity Check

Stationarity Checks

70/30 Split

ADF Statistic: -0.6655393420584818
P-value: 0.85547487720912
Critical Values: {'1%': np.float64(-3.432889109077145), '5%': np.float64(-2.862661779822738), '10%': np.float64(-2.5673671643151064)}

Diff'd ADF Statistics: -16.775679266673862
P-Value: 0.00000
Critical Values: {'1%': np.float64(-3.432889109077145), '5%': np.float64(-2.862661779822738), '10%': np.float64(-2.5673671643151064)}



DATA ANALYSIS PROCESS

```
Performing stepwise search to minimize aic
ARIMA(2,1,2)(0,0,0)[0] intercept : AIC=14842.445, Time=0.67 sec
ARIMA(0,1,0)(0,0,0)[0] intercept : AIC=14840.486, Time=0.03 sec
ARIMA(1,1,0)(0,0,0)[0] intercept : AIC=14838.307, Time=0.09 sec
ARIMA(0,1,1)(0,0,0)[0] intercept : AIC=14838.167, Time=0.15 sec
ARIMA(0,1,0)(0,0,0)[0] intercept : AIC=14841.407, Time=0.03 sec
ARIMA(1,1,1)(0,0,0)[0] intercept : AIC=14838.458, Time=0.39 sec
ARIMA(0,1,2)(0,0,0)[0] intercept : AIC=14839.472, Time=0.29 sec
ARIMA(1,1,2)(0,0,0)[0] intercept : AIC=14840.411, Time=0.69 sec
ARIMA(0,1,1)(0,0,0)[0] intercept : AIC=14839.351, Time=0.06 sec
```

Best model: ARIMA(0,1,1)(0,0,0)[0] intercept
Total fit time: 2.403 seconds

SARIMAX Results

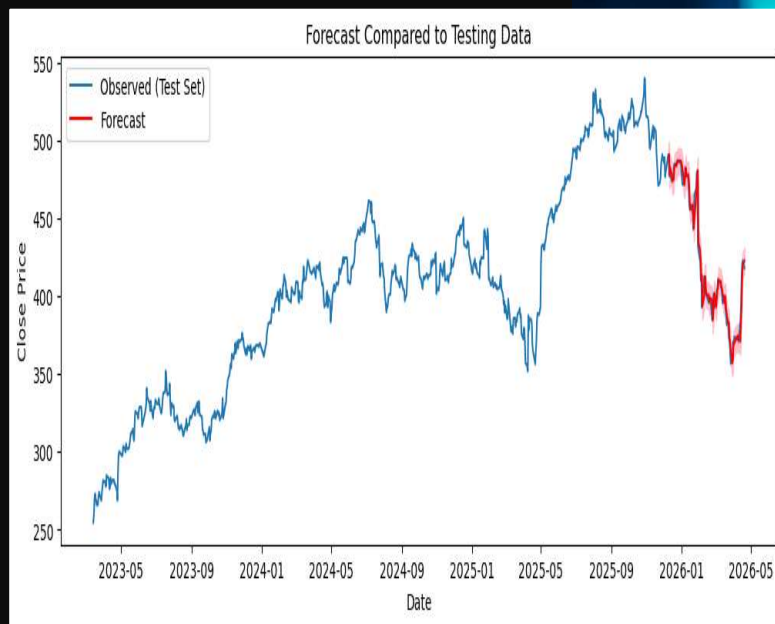
Dep. Variable:	y	No. Observations:	2588			
Model:	SARIMAX(0, 1, 1)	Log Likelihood	-7416.084			
Date:	Tue, 15 Sep 2026	AIC	14838.167			
Time:	12:53:41	BIC	14855.742			
Sample:	0	HQIC	14844.537			
	- 2588					
Covariance Type:	opg					
	coef	std err	z	P> z	[0.025	0.975]
intercept	0.1432	0.081	1.767	0.077	-0.016	0.302
ma.L1	-0.0416	0.014	-2.925	0.003	-0.069	-0.014
sigma2	18.0920	0.191	94.854	0.000	17.718	18.466
Ljung-Box (L1) (Q):	0.00	Jarque-Bera (JB):	16552.16			
Prob(Q):	0.97	Prob(JB):	0.00			
Heteroskedasticity (H):	26.23	Skew:	-0.53			
Prob(H) (two-sided):	0.00	Kurtosis:	15.35			

Model Identification

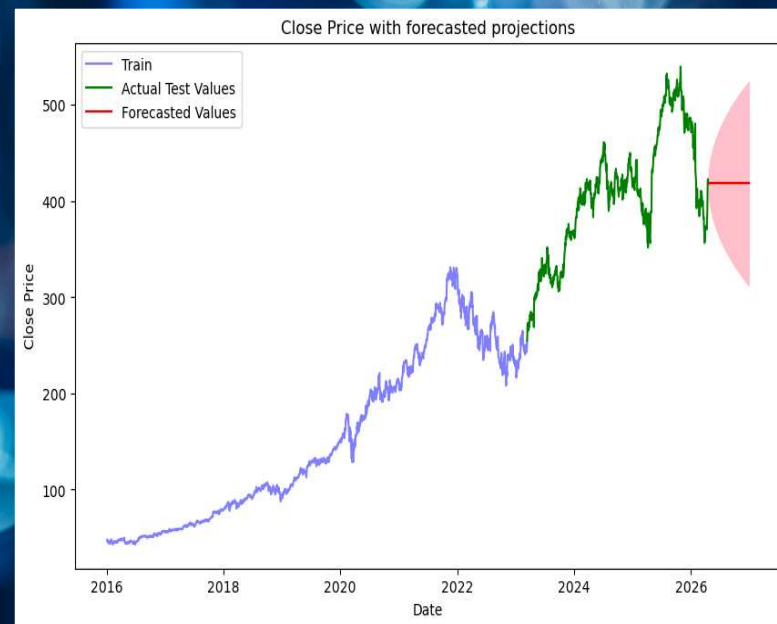
- Auto_arima function

DATA ANALYSIS PROCESS

Forecasting



Forecast compared to Test Data



Close Price with Forecasted Projections



OUTLINE OF FINDINGS

- Observed Range
 - January 2016 to April 2026
- Forecast Range
 - 6 months
- Model Accuracy
 - Calculated MAPE=1.2%
 - 98.8% Accuracy
 - Rejection of Null Hypothesis

DATA ANALYSIS PROCESS

Limitations

Data

- Accuracy and completeness of publicly available data

Tools

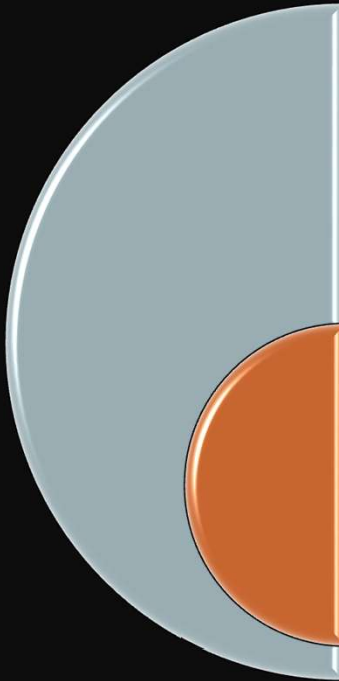
- Jupyter Notebook
- Base Environment not starting (Minhas, 2025)
- Matplotlib
- Visualizations are static

Techniques

- ARIMA
- Not for long term forecasting (Hayes, 2026)
- Only uses historical data
- Auto_arima
- Don't get to see how well other models perform

DATA ANALYSIS PROCESS

Proposed Actions



Model Improvement

- Parameter Changes
- Change Split

Alternative Models

- LSTM
- Capture complex patterns of market data (B, 2025)



BENEFITS OF THE STUDY

- Model Accuracy
 - Calculated MAPE=1.2%
 - 98.8% accuracy
- Not limited to Microsoft
 - Since the model is based off of historical stock data, it can be used to forecast the future close prices of other companies as well.



SOURCES

- B, A. (2025, March 21). Predicting stock prices using lstms: A step-by-step guide to time series forecasting | by aditi B | medium. Medium. <https://medium.com/@aditib259/predicting-stock-prices-using-lstms-time-series-forecasting-a-step-by-step-guide-a70ebb04bbb8>
- Hayes, A. (2026, July 17). Master Arima: Your guide to time series forecasting. Investopedia. <https://www.investopedia.com/terms/a/autoregressive-integrated-moving-average-arima.asp>
- Minhas, M. S. (2025, January 18). 10 reasons why I stopped using jupyter lab. Towards Data Science. <https://towardsdatascience.com/10-reasons-why-i-stopped-using-jupyter-lab-498e958424ac/>